

TOLUCA INTL, Mexico

WMO: 766753

Lat: 19.337N

Lon: 99.566W

Elev: 8466

StdP: 10.72

Time Zone: -6.00 (MEX)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.4	30.6	14.6	15.8	64.5	17.9	18.6	60.9	21.1	60.2	18.9	61.2	0.8	140	0.485

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	28.5	79.1	53.9	77.1	53.4	75.2	52.8	60.4	69.7	59.2	68.7	58.6	68.0	9.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.5	97.2	62.5	56.8	94.8	62.1	55.5	90.4	61.1	31.0	69.7	30.2	68.5	29.5	68.5	68.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	16.5	14.3	22.7	83.3	3.1	2.2	20.5	84.9	18.6	86.2	16.9	87.4	14.6	88.9		
(5)			21.8	62.8	2.9	2.3	19.8	64.5	18.1	65.8	16.5	67.1	14.4	68.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.4	49.7	52.9	55.6	58.8	60.8	60.5	59.1	59.3	59.4	57.2	53.4	50.4		
	DBStd	4.65	2.74	2.97	3.00	2.97	2.78	2.37	1.87	1.82	2.08	3.15	3.37	3.04		
	HDD50	90	38	9	2	0	0	0	0	0	0	1	9	31		
	HDD65	3133	475	340	293	187	132	137	182	176	169	243	347	452		
	CDD50	2438	29	88	175	264	336	314	283	289	281	223	111	45		
	CDD65	4	0	0	0	0	3	1	0	0	0	0	0	0		
	CDH74	689	4	28	97	223	246	72	4	3	2	4	4	2		
	CDH80	45	0	1	2	14	25	3	0	0	0	0	0	0		
(14) Wind	WSAvg	4.6	4.2	4.6	5.0	5.0	4.9	4.8	4.4	4.7	4.7	4.6	4.1	3.8		
(15) Precipitation	PrecAvg	41.4	0.8	0.7	0.6	1.3	3.3	7.3	8.6	8.1	7.0	2.6	0.9	0.4		
	PrecMax	49.1	3.7	5.1	2.1	3.1	8.0	11.9	15.0	11.5	13.5	4.6	3.3	2.1		
	PrecMin	35.2	0.1	0.0	0.0	0.2	0.2	2.2	5.9	5.3	3.6	0.1	0.1	0.0		
	PrecStd	4.3	0.7	1.0	0.6	0.8	1.8	2.0	1.7	1.5	2.3	1.3	0.7	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.3	77.2	79.2	82.1	82.7	79.1	73.8	73.7	73.6	73.8	73.8	73.4		
		MCWB	49.8	50.6	52.5	55.1	55.9	54.5	56.3	56.6	55.7	54.2	52.7	51.4		
	2%	DB	70.1	73.8	76.8	78.9	79.1	75.6	71.8	71.8	71.6	71.9	71.7	70.7		
		MCWB	48.9	50.3	51.9	53.7	54.6	54.5	56.6	56.5	56.1	54.0	52.2	50.2		
	5%	DB	68.1	71.7	74.8	76.8	76.9	73.3	69.9	69.9	69.7	70.0	69.8	68.3		
		MCWB	48.8	50.0	51.0	53.2	54.1	54.7	56.4	56.3	55.8	54.0	51.8	49.5		
	10%	DB	66.0	69.6	71.8	73.7	73.7	70.1	67.9	68.0	67.8	68.0	67.7	66.2		
		MCWB	48.4	49.8	50.2	51.9	53.7	55.9	56.3	56.2	56.0	54.0	51.3	49.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.1	56.4	57.5	60.2	61.0	61.4	61.1	60.9	60.9	59.9	57.7	55.7		
		MCDB	66.5	70.4	72.0	74.8	73.4	70.9	68.9	68.5	68.2	68.3	68.5	68.4		
	2%	WB	52.1	53.9	55.2	57.5	59.1	59.4	59.2	59.2	59.2	58.4	56.1	53.7		
		MCDB	65.3	68.8	71.3	73.4	72.6	68.7	67.8	67.5	66.6	66.6	66.6	66.1		
	5%	WB	50.7	52.2	53.5	55.8	57.7	58.7	58.3	58.3	58.3	57.1	54.8	52.0		
		MCDB	64.0	66.7	70.1	71.4	70.8	67.7	66.8	66.6	65.7	65.6	64.9	63.9		
	10%	WB	49.5	50.8	51.8	54.2	56.2	57.5	57.0	57.2	57.2	55.9	53.4	50.4		
		MCDB	62.8	65.1	68.0	69.2	68.5	66.1	65.3	65.4	64.7	64.1	63.4	62.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	34.3	34.8	34.7	32.7	28.5	21.7	20.4	20.2	19.6	24.4	30.6	34.0		
		MCDBR	38.3	38.5	38.3	36.2	33.0	27.3	23.4	23.0	22.5	27.6	33.5	37.0		
		MCWBR	19.5	18.3	17.2	16.3	14.0	11.7	11.3	10.8	10.5	13.3	16.8	18.9		
	5% WB	MCDBR	34.8	34.8	34.8	32.9	28.6	21.8	21.0	20.6	19.6	22.8	28.6	33.1		
		MCWBR	19.6	18.7	18.0	16.9	13.9	11.4	11.3	10.7	10.1	12.0	15.7	18.7		
(40) Clear-Sky Solar Irradiance	taub		0.377	0.394	0.401	0.469	0.429	0.481	0.464	0.446	0.349	0.408	0.327	0.367		
	taud		2.202	2.144	2.145	1.982	2.143	1.975	2.045	2.105	2.449	2.190	2.445	2.231		
	Ebn at Noon		284	285	287	268	276	259	264	271	299	278	300	283		
	Edh at Noon		43	48	49	59	50	58	54	51	36	45	33	40		
(44) All-Sky Solar Radiation	RadAvg		1565	1827	2048	2103	1943	1771	1767	1764	1574	1629	1584	1540		
	RadStd		103	151	118	80	122	127	148	98	114	118	119	86		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	+1.26	-3.35	N/A	-1.69	-2.36	-54	N/A	N/A	N/A	
(47) Regional (7 neighbors)		+0.67	+1.89	-3.15	N/A	N/A	N/A	-4	-229	+297	+5	

Nomenclature: See separate page